

AML TRANSACTION MONITORING AND RISK ANALYTICS:

TECHNICAL CASE STUDY

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TOOLS:

- **PYTHON**
- **PANDAS**
- **NUMPY**
- **SCIKIT-LEARN**
- **XG BOOST**
- **POWER BI**
- **DAX**

DATE: JULY 2026

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The increasing volume of financial transactions has made it more difficult for financial institutions to identify suspicious activities using manual reviews or traditional monitoring methods alone. This project explores how data analytics can be used to support Anti-Money Laundering (AML) transaction monitoring by transforming raw transaction data into meaningful information that helps identify suspicious behaviour and assess account risk.

The project involved preparing and analysing transaction data, creating behavioural features that provide a better understanding of how accounts interact, identifying suspicious transaction patterns through network analysis, and developing a risk scoring framework to prioritize accounts that may require further investigation. The engineered features were also used to build and evaluate machine learning models for account risk classification.

The analytical results were presented in an interactive Power BI dashboard, providing a clear view of transaction activity, risk distribution, suspicious transaction patterns, and high-risk accounts.

Overall, this project demonstrates how data can be transformed into actionable insights that support more effective transaction monitoring, improve risk assessment, and assist financial institutions in making informed decisions during AML investigations. This project was developed to design and implement an intelligent Anti Money Laundering monitoring system capable of identifying suspicious account behaviour from large scale financial transaction data.

Unlike traditional transaction reporting, the solution focuses on account level risk assessment by analysing how customers behave over time rather than evaluating transactions in isolation. Historical transaction data was transformed into behavioural profiles that capture spending patterns, fund movement, network relationships, and indicators commonly associated with money laundering.

Anti-Money Laundering (AML) is an essential part of protecting the financial system from the movement of illicit funds. Financial institutions are required to monitor large volumes of transactions, identify unusual activities, and investigate accounts that may present a higher risk of financial crime. As transaction volumes continue to increase, data analytics has become an important tool for supporting these activities.

This project focuses on analysing transaction data to better understand account behaviour and transaction relationships. Instead of relying only on the information contained in individual transactions, the project creates additional information from the data that provides a clearer picture of how accounts behave and interact. These insights are then used to identify suspicious patterns, assess account risk, and support AML monitoring through data-driven analysis.

The objectives of this project include:

- * Prepare and clean transaction data to ensure it is suitable for analysis.**
- * Create behavioural features that provide deeper insights into account activity and transaction relationships.**
- * Explore transaction data to identify patterns, trends, and unusual activities.**
- * Detect suspicious transaction patterns using fan-in and fan-out network analysis.**
- * Develop a risk scoring framework to classify accounts according to their level of risk.**
- * Build and evaluate machine learning models for account risk classification.**
- * Develop an interactive Power BI dashboard to communicate key findings and support transaction monitoring.**
- * Demonstrate how data analytics can improve AML risk assessment and support better decision-making.**

Financial institutions process thousands of transactions every day, making it difficult for compliance teams to manually review every transaction or identify suspicious activities in a timely manner. Traditional rule-based monitoring systems can generate large numbers of alerts, many of which are false positives, while more complex transaction patterns may go unnoticed.

In many cases, a single transaction provides limited information about whether an account is genuinely suspicious. A clearer understanding is gained by analysing how accounts behave over time, how they interact with other accounts, and how funds move across the transaction network.

This project addresses this challenge by transforming raw transaction data into behavioural features that support deeper analysis of account activity. By combining transaction analysis, network analytics, risk scoring, and machine learning, the project demonstrates a data-driven approach to identifying higher-risk accounts and supporting more efficient AML investigations.

To improve detection and investigation, financial institutions require intelligent monitoring systems that analyse customer behaviour, identify suspicious transaction patterns, and prioritize high risk accounts using data driven techniques.

This project is based on the IBM Transactions for Anti Money Laundering (HI Small) dataset, a synthetic financial transaction dataset developed by IBM Research and published on Kaggle by Erik Altman. The dataset was designed to simulate realistic banking transactions while incorporating labelled money laundering activities, making it well suited for developing and evaluating Anti Money Laundering (AML) detection solutions.

The primary transaction dataset contains 5,078,345 financial transactions involving approximately 515,000 bank accounts over a ten-day period. Each transaction record includes information such as the sending and receiving institutions, participating accounts, transaction amount, payment details, timestamps, and a laundering label indicating whether the transaction is associated with suspicious activity.

To support account-level analysis, the transaction data was aggregated and transformed into an engineered dataset containing approximately 497,000 unique account profiles. This engineered dataset formed the foundation for feature engineering, network analytics, custom risk scoring, machine learning, and executive dashboard reporting.

In addition to the transaction dataset, the HI Small Patterns dataset was used to analyse money laundering typologies. This dataset was processed independently and served as the basis for the AML Transaction Pattern Analysis dashboard, providing insights into laundering patterns, transaction behaviour, and investigative trends.

Dataset Source:

- * Dataset: IBM Transactions for Anti Money Laundering (HI Small)**
- * Pattern Dataset: HI Small Patterns**

* Developed by: IBM Research

* Published on Kaggle by: Erik Altman

KEY VARIABLES

S/N	VARIABLE	DESCRIPTION
1.	Timestamp	Date and time the transaction occurred
2.	From Bank	Financial institution initiating the transaction
3.	Account	Unique identifier of the sending account
4.	To Bank	Financial institution receiving the transaction
5.	Account.1	Unique identifier of the receiving account
6.	Amount Paid	Amount debited from the sender's account
7.	Payment Currency	Currency used by the sender
8.	Amount Received	Amount credited to the receiving account
9.	Receiving Currency	Currency received by the beneficiary
10.	Payment Format	Transaction channel or payment method (e.g., ACH, Wire, Cheque, Credit card)
11.	Is Laundering	Binary target variable indicating whether the transaction is associated with money laundering (1= YES, 0= NO)

Table.1: Key Variables, IBM AML Dataset

The proposed solution was designed as an end-to-end Anti Money Laundering (AML) analytics framework that converts large-scale transaction data into actionable risk intelligence. The architecture follows a structured workflow, beginning with raw transaction data and progressing through data preparation, behavioural analysis, risk scoring, predictive modelling, and interactive reporting.

Two datasets were used throughout the project. The IBM HI Small Transactions dataset served as the primary source for account-level analysis, feature engineering, and machine learning, while the HI Small Patterns dataset was analysed separately to provide insights into common money laundering typologies and transaction behaviours.

After data preparation, transaction records were aggregated into account-level profiles and enriched with behavioural features such as Fan In, Fan Out, Transaction Velocity, Active Days, Laundering Rate, Total Sent, Total Received, Risk Score, Risk Category, and Risk Driver. These engineered features formed the foundation for network analytics and the custom risk scoring framework.

The engineered dataset was subsequently used to train and evaluate machine learning models for predicting suspicious accounts. The final outputs were presented through a two-page Power BI report consisting of an Executive Overview of AML Risk Monitoring dashboard for account-level risk assessment and an AML Transaction Pattern Analysis dashboard for analysing laundering patterns using the HI Small Patterns dataset.

The overall architecture provides a scalable framework for detecting suspicious financial behaviour, prioritising high-risk accounts, and supporting AML investigations through explainable analytics and interactive business intelligence.

**IBM AML HI Small Transactions Dataset
(5+ Million Transactions)**



Data Preparation & Cleaning



Account-Level Aggregation



Feature Engineering



Network Analytics (Fan In, Fan Out, Velocity)



Custom Risk Scoring Framework



Machine Learning Model Development



Engineered AML Account Dataset



Executive Overview of AML Risk Monitoring Dashboard

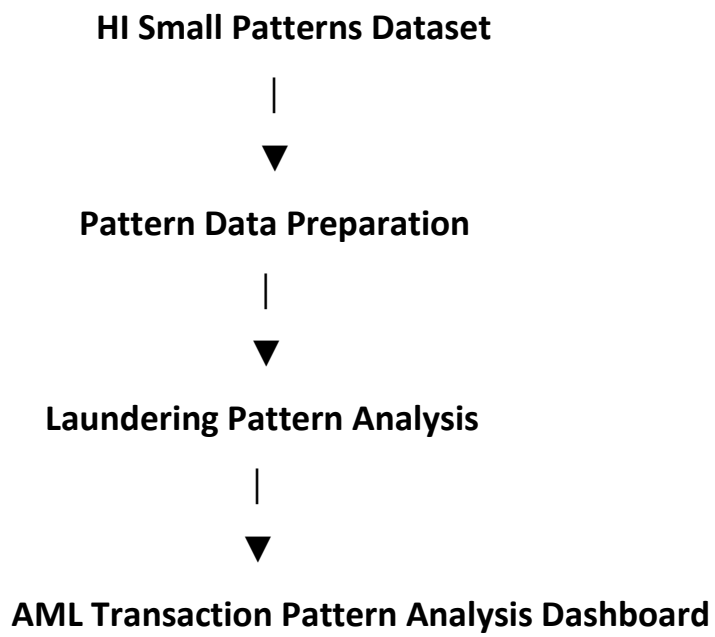


Fig.1: End-to-End AML Solution Architecture

High-quality data is essential for reliable risk analysis and predictive modelling. Before any analytical or machine learning tasks were performed, the transaction dataset underwent a structured data preparation process to improve data quality, ensure consistency, and create a reliable foundation for feature engineering.

The transaction dataset was imported into Google Colab using the Pandas library, where an initial assessment was conducted to understand its structure, identify data quality issues, and verify the integrity of each variable. This included reviewing data types, checking for missing values, validating transaction records, and examining the distribution of key variables.

To prepare the data for analysis, timestamp values were converted into a standard datetime format, enabling the calculation of time-based behavioural features such as Active Days, First Transaction, Last Transaction, and Transaction Velocity. Numeric fields representing transaction amounts were also validated to ensure they could be used for statistical analysis and feature engineering.

Rather than analysing individual transactions, the cleaned transaction records were aggregated at the account level. This transformation produced a single behavioural profile for each account by consolidating historical transaction activity into meaningful analytical measures. The resulting account-level dataset became the foundation for behavioural analysis, risk scoring, machine learning, and dashboard development.

The HI Small Patterns dataset was prepared independently for transaction pattern analysis. Since its purpose was to provide insight into laundering typologies rather than account behaviour, it remained separate from the engineered AML dataset and was used exclusively in the AML Transaction Pattern Analysis dashboard.

Following data preparation, the project produced two structured analytical datasets:

- * An engineered AML account dataset containing behavioural and risk-related features for account-level analysis.**
- * A prepared pattern dataset supporting the analysis and visualization of laundering typologies.**

This structured approach ensured that subsequent analyses were performed on clean, consistent, and analysis-ready data while maintaining a clear separation between account risk assessment and laundering pattern intelligence.

Feature engineering was performed to transform transaction-level records into meaningful account-level behavioural profiles.

Rather than analysing individual transactions in isolation, the engineered features summarise historical account activity, transaction behaviour, and network relationships, providing a stronger foundation for risk assessment and predictive modelling.

The transaction data was aggregated by sender account, resulting in a single analytical record for each account. The engineered features were grouped into four categories based on their analytical purpose.

Transaction Activity Features:

These features describe the overall transaction behaviour of each account.

S/N	FEATURE	DESCRIPTION
1.	Transaction Count	Total number of transactions initiated by the account
2.	Total Sent	Value of funds transferred by the account
3.	Total Received	Value of funds received by the account
4.	Unique Receivers	Number of distinct recipient accounts
5.	First Txn	Date of account's first recorded transaction
6,	Last Txn	Date of account's most recent transaction
7.	Active Days	Number of days between the first and last recorded transaction

Table.2.1: Transaction Activity Features

Network Analytics Features:

These features measure how accounts interact within the transaction network.

S/N	FEATURE	DESCRIPTION
1.	Fan-in	Number of incoming relationships associated with the account
2.	Fan-out	Number of outgoing relationships associated with the account
3.	Fan-in Percentile	Relative ranking based on Fan in
4.	Fan-out Percentile	Relative ranking based on Fan out

Table.2.2: Network Analytics Features

Behavioural Features:

These standardise account activity, allowing comparisons across accounts with different transaction volumes.

S/N	FEATURE	DESCRIPTION
1.	TXN Percentile	Relative ranking based on transaction count
2.	Sent Percentile	Relative ranking based on total amount sent
3.	TXN Log	Log-transformed transaction count used to reduce skewness
4.	Sent Log	Log-transformed total amount sent to reduce skewness

Table.2.3: Behavioural Features

Risk Features:

These features quantify historical laundering activity, and the overall risk associated with each account.

S/N	FEATURE	DESCRIPTION
1.	Laundering Count	Number of confirmed laundering transactions linked to the account
2.	Laundering Rate	Percentage of transactions associated with money laundering
3.	Risk Score	Composite score calculated from multiple behavioural indicators
4.	Risk Category	Final classification of each account into Low, Medium, High, or Critical
5.	Risk Driver	Behavioural metrics contributing most significantly to the account's risk score

Table.2.4: Risk Features

Feature Engineering Outcome:

The feature engineering process transformed more than 5 million transaction records into approximately 497,000 account level profiles, creating a structured dataset for network analytics, risk scoring, machine learning, and business intelligence.

The engineered features not only quantified transaction behaviour, but also provided explainable indicators, that helped distinguish normal account activity from behaviour commonly associated with money laundering.

Following feature engineering and the development of the account level risk scoring framework, exploratory data analysis was carried out to better understand the characteristics of the engineered dataset and validate the behavioural patterns captured by the newly created features. Rather than analysing raw transactions, the analysis focused on account level behaviour after each account had been assigned a risk score, risk category and primary risk indicator.

The objective was to identify behavioural trends, examine relationships between engineered variables and confirm that the extracted features meaningfully differentiated low risk accounts from suspicious ones before machine learning modelling and dashboard development.

The analysis focused on:

- * Distribution of risk scores across monitored accounts.
- * Distribution of transaction counts to understand customer activity levels.
- * Distribution of total funds sent to identify accounts with unusually large transaction volumes.
- * Distribution of Fan-In and Fan-Out values to examine the concentration and dispersion of fund flows.
- * Correlation analysis between engineered features using a correlation heatmap.

The exploratory analysis showed that most accounts exhibited normal transaction behaviour and were classified as Low Risk, while a relatively small proportion demonstrated behavioural characteristics associated with elevated financial crime risk. Strong positive relationships were observed between several engineered variables, particularly transaction activity, network connectivity and percentile based behavioural features, confirming that these variables captured different aspects of account behaviour rather than duplicating the same information.

The analysis also highlighted the highly skewed nature of financial transaction data, where a small number of accounts were responsible for exceptionally large transaction volumes and extensive fund movement. These observations reinforced the effectiveness of the engineered features and provided confidence that the dataset was suitable for both predictive modelling and business intelligence reporting.

Traditional rule-based AML systems often struggle to detect coordinated laundering activities because suspicious accounts rarely operate in isolation. To address this limitation, network based behavioural analysis was incorporated into the project to identify how accounts interacted across the transaction ecosystem.

Instead of analysing individual transactions, relationships between accounts were aggregated to the account level to reveal behavioural patterns that are commonly associated with money laundering.

The following network features were engineered:

- * **Fan Out** – Number of unique recipient accounts receiving funds from a sender.
- * **Fan In** – Number of unique accounts sending funds into an account.
- * **Unique Receivers** – Total distinct beneficiaries linked to each sender.
- * **Transaction Count** – Overall transaction activity generated by each account.
- * **Total Sent and Total Received** – Monetary flow into and out of each account.
- * **Active Days** – Number of days an account remained active during the observation period.

These behavioural indicators provided a broader understanding of account relationships beyond transaction values alone. Accounts exhibiting unusually high transaction volumes, extensive fund dispersion, or concentrated incoming activity were more likely to demonstrate suspicious financial behaviour.

The engineered network features became the foundation of the risk scoring framework and later served as important predictors during machine learning model development.

Detecting suspicious financial activity requires more than analysing individual transactions. Accounts involved in money laundering often display a combination of behavioural characteristics that, when considered together, provide a clearer indication of risk than any single metric alone.

To capture this behaviour, a custom risk scoring framework was developed to evaluate every account using multiple engineered features derived from the transaction data. Instead of relying on fixed thresholds, the framework combined transaction activity, network behaviour and historical laundering indicators into a single composite Risk Score that reflects the overall behavioural profile of each account.

The scoring framework considered the following behavioural indicators:

- * Transaction Count
- * Total Sent
- * Total Received
- * Unique Receivers
- * Fan In
- * Fan Out
- * Fan In Percentile
- * Fan Out Percentile
- * Transaction Percentile
- * Sent Percentile
- * Laundering Count
- * Laundering Rate
- * Active Days

Each feature contributed to the final score based on predefined business rules designed to highlight behaviours commonly associated with money

laundering. Accounts demonstrating unusually high transaction activity, extensive fund movement, elevated network connectivity or a higher proportion of historical laundering transactions accumulated higher overall risk scores.

Once the composite score had been calculated, each account was assigned to one of four operational risk categories:

S/N	RISK CATEGORY	PURPOSE
1.	Low	Behaviour consistent with normal account activity
2.	Medium	Accounts exhibiting moderate behavioural anomalies that may require periodic monitoring
3,	High	Accounts displaying multiple suspicious indicators requiring further investigation
4.	Critical	Accounts demonstrating strong behavioural characteristics associated with potential money laundering and requiring immediate attention

Table.3: Risk Category

Unlike traditional rule-based monitoring, this framework evaluates customer behaviour from multiple perspectives simultaneously, providing a more balanced and explainable assessment of account risk. By combining transaction behaviour, network relationships and historical laundering activity into a single score, the framework enables investigators to prioritise accounts based on overall behavioural risk rather than isolated events.

The resulting Risk Score and Risk Category formed the foundation for the subsequent machine learning model and the executive risk monitoring dashboard, ensuring consistency between predictive modelling and business reporting.

Following feature engineering, network analytics, the development of the custom risk scoring framework, and exploratory data analysis, the engineered account-level dataset was used to develop predictive machine learning models capable of identifying suspicious accounts based on their behavioural characteristics.

Unlike the rule-based risk scoring framework, which assigns risk according to predefined business rules, the machine learning models were designed to learn patterns directly from historical account behaviour. The objective was to determine whether the engineered behavioural features could accurately distinguish accounts exhibiting money laundering characteristics from normal customer activity while providing an additional layer of intelligence for AML monitoring.

This stage transformed the project from descriptive and diagnostic analytics into predictive analytics by evaluating the ability of machine learning to classify account risk using behavioural and network-based indicators.

Feature Selection:

The models were trained using the engineered account-level dataset created during the feature engineering stage. Each record represented a unique customer account summarising its transactional behaviour over the observation period.

The predictor variables included behavioural, transactional and network-based features engineered from the original transaction data, including:

- * Transaction Count
- * Total Sent
- * Total Received
- * Unique Receivers
- * Fan In
- * Fan Out

- * Fan In Percentile**
- * Fan Out Percentile**
- * Transaction Percentile**
- * Sent Percentile**
- * Active Days**
- * Laundering Count**
- * Laundering Rate**
- * Transaction Log (txn_log)**
- * Sent Log (sent_log)**

These variables were selected because they capture different dimensions of customer behaviour, including transaction frequency, fund movement, account connectivity and historical exposure to suspicious activity. Together, they provide a comprehensive behavioural profile that is more informative than relying on individual transactions alone.

The target variable for model training was the Risk Category, where each account had already been classified as Low, Medium, High, or Critical risk by the custom risk scoring framework.

Training and Testing Strategy:

To evaluate model performance objectively, the engineered dataset was divided into training and testing datasets using an 80:20 split.

The training dataset was used to learn behavioural patterns from historical account activity, while the testing dataset remained unseen during training and was used to assess each model's ability to classify new accounts accurately. This approach reduced the likelihood of overfitting and provided a realistic measure of how the models would perform on previously unseen data.

Addressing Class Imbalance:

One of the most significant challenges encountered during model development was the highly imbalanced distribution of account risk.

The overwhelming majority of accounts belonged to the Low-Risk category, while High and Critical Risk accounts represented only a small proportion of the dataset. This imbalance is common in Anti-Money Laundering datasets, where genuinely suspicious accounts are significantly outnumbered by legitimate customer activity.

Training a model directly on this distribution produced misleading results. Although the baseline model achieved extremely high overall accuracy, it largely learned to predict the majority class and failed to identify many suspicious accounts. This highlighted the limitations of relying on accuracy alone when evaluating AML classification models.

To improve the model's ability to recognise minority classes, the Synthetic Minority Oversampling Technique (SMOTE) was applied to the training dataset.

Rather than duplicating existing observations, SMOTE generates synthetic examples of minority classes by interpolating between similar records. This creates a more balanced training dataset while preserving the underlying behavioural characteristics of suspicious accounts.

Applying SMOTE allowed the models to learn more representative decision boundaries, improving their ability to recognise High and Critical Risk accounts without altering the testing dataset.

Model Development:

To identify the most suitable predictive model, four classification approaches were developed and evaluated.

The modelling process progressed through several stages, with each model addressing limitations identified in the previous approach.

The evaluated models were:

- * Baseline Random Forest
- * Random Forest with SMOTE
- * Tuned Random Forest
- * XGBoost with SMOTE

The Baseline Random Forest established an initial benchmark using the original class distribution. A second model incorporated SMOTE to improve learning from minority risk categories. The Random Forest model was then further refined through hyperparameter tuning to improve its ability to generalise. Finally, an XGBoost classifier was trained using the balanced dataset to determine whether gradient boosting could provide stronger predictive performance than Random Forest.

Using multiple models enabled a systematic comparison of different modelling strategies while demonstrating the impact of class balancing and model optimisation on AML classification performance.

	Model	F1 Score	Precision	Recall	Accuracy
0	Baseline Random Forest	0.0000	0.000	0.0000	0.9990
1	SMOTE + Random Forest	0.0261	0.013	0.6814	0.9482
2	Tuned Random Forest	0.0240	0.012	0.7794	0.9346
3	SMOTE + XGBoost	0.0278	0.014	0.7206	0.9487

Fig.2: Model Performance Comparison

Comparative Analysis:

The comparison of model performance demonstrated that addressing class imbalance had a significant impact on the ability to identify suspicious accounts.

The Baseline Random Forest achieved an overall accuracy of 99.9%, but produced zero precision, zero recall and an F1-score of zero because it overwhelmingly predicted the majority class. Although the accuracy appeared excellent, the model failed to identify laundering accounts and therefore provided little practical value for AML investigations.

Introducing SMOTE substantially improved minority-class detection. The Random Forest model trained on the balanced dataset achieved a recall of 68.14%, indicating a significant improvement in its ability to identify suspicious accounts.

Further optimisation through hyperparameter tuning increased recall to 77.94%, demonstrating greater sensitivity to high-risk accounts. However, this improvement came with a slight reduction in precision, illustrating the trade-off between identifying more suspicious accounts and increasing false positives.

The SMOTE + XGBoost model delivered the strongest overall balance across the evaluation metrics. While its overall accuracy (94.87%) was lower than the baseline model, it achieved the highest F1-score (0.0278) and Precision (0.014) while maintaining a strong Recall (72.06%). These results indicate a better balance between identifying suspicious accounts and limiting unnecessary investigations.

Final Model Selection:

Based on the comparative evaluation, the SMOTE + XGBoost model was selected as the final predictive model for this project.

The selection was based on its overall balance of Precision, Recall and F1-score rather than accuracy alone. In Anti-Money Laundering applications, correctly identifying suspicious accounts is considerably more important than achieving high overall accuracy by predominantly predicting legitimate accounts.

By combining the engineered behavioural features, network analytics, custom risk scoring framework and SMOTE-balanced training data, the final model demonstrated a stronger ability to identify high-risk accounts within a highly imbalanced financial dataset.

Having identified the most suitable predictive model, the next stage evaluates its performance in greater detail using classification metrics and discusses its effectiveness for supporting risk-based AML monitoring and investigation.

The final phase of this project focused on translating the analytical outputs generated in Python into an interactive business intelligence solution using Microsoft Power BI. While Python was used to engineer features, calculate behavioural risk metrics, develop the risk scoring framework and build predictive models, Power BI transformed those outputs into intuitive dashboards that support operational monitoring and decision-making.

Rather than presenting raw analytical results, the dashboards were designed to answer the key questions AML investigators and compliance teams ask every day:

- * Which accounts require immediate attention?
- * What behaviours are driving risk?
- * Which laundering patterns are most common?
- * Where should investigators focus first?

To achieve this, the final solution was organised into two complementary dashboard pages.

1. Executive Overview of AML Risk Monitoring:

The Executive Overview dashboard provides a high-level summary of the overall AML risk landscape across all monitored accounts. It is intended for compliance managers, AML analysts and decision makers who require a quick understanding of institutional risk exposure without reviewing individual transactions.

The dashboard combines the engineered account-level dataset, the custom risk scoring framework and additional business logic implemented within Power BI to present a consolidated view of suspicious financial activity.

At the top of the dashboard, key performance indicators summarise the overall monitoring environment, including:

- * Total Transaction Value**
- * Total Accounts Monitored**
- * High & Critical Risk Accounts**
- * Confirmed Laundering Accounts**
- * Average Risk Score**

These KPIs provide an immediate assessment of transaction volume, overall customer exposure and the concentration of high-risk accounts.

The dashboard then examines the behavioural drivers contributing to elevated risk through the Primary Risk Indicator classification. Unlike the behavioural features engineered in Python, the Primary Risk Indicator was created within Power BI to identify the single dominant factor responsible for each account's risk classification. This allows investigators to move beyond numerical risk scores and quickly understand the primary behaviour responsible for an account being flagged.

To support prioritisation, accounts are also grouped by Risk Category, enabling compliance teams to distinguish between Low, Medium, High and Critical risk populations and allocate investigative resources accordingly.

A detailed investigation table lists the highest-risk accounts together with their associated bank, account number, transaction volume, total funds sent and received, number of unique receivers, risk score and primary behavioural indicator. This enables analysts to move seamlessly from executive-level monitoring into individual account investigations.

The dashboard concludes with a business insight panel that summarises the major findings derived from the analysis. Rather than requiring users to interpret multiple visuals independently, the insight section provides a concise narrative explaining the observed concentration of suspicious behaviour and how the integrated risk scoring framework supports risk-based AML investigations.

Overall, this dashboard serves as an executive monitoring solution that combines descriptive analytics with actionable business intelligence for ongoing financial crime surveillance.

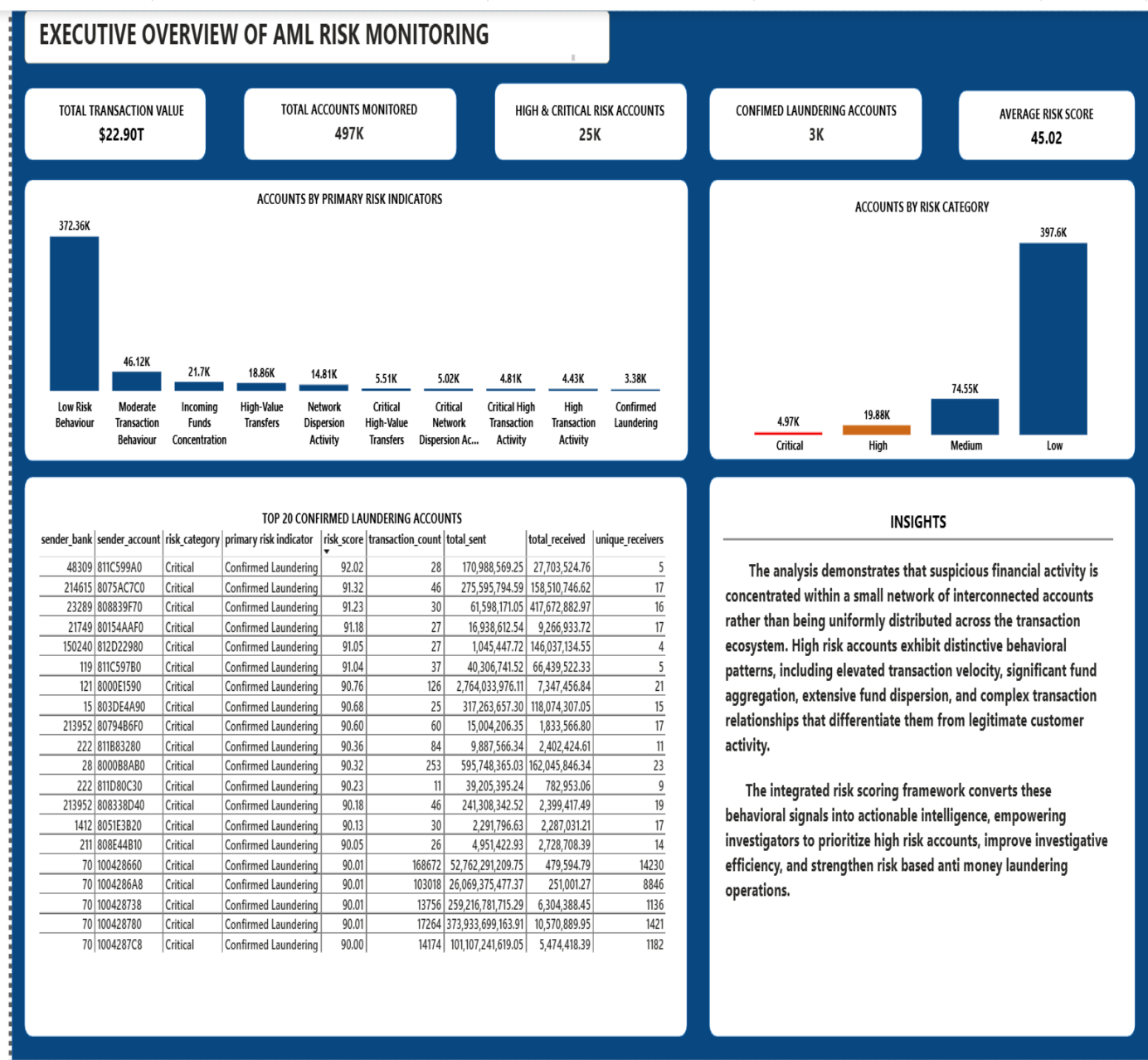


Fig.3: Executive Overview of AML Risk Monitoring Dashboard

2. AML Transaction Pattern Analysis:

While the Executive Overview dashboard focuses on individual account risk, the second dashboard shifts attention to suspicious transaction networks by analysing common money laundering patterns detected across the dataset.

This dashboard provides investigators with a structured view of how suspicious transactions are organised, allowing them to understand not only which accounts are risky but also how laundering activities are being executed.

The dashboard begins with four summary indicators that describe the overall characteristics of detected laundering patterns:

- * Total Detected Laundering Patterns**
- * Average Pattern Duration**
- * Average Transactions per Pattern**
- * Longest Pattern Duration**

These metrics provide an overview of the scale and operational complexity of suspicious transaction networks identified during the analysis.

The central section of the dashboard allows investigators to explore multiple laundering typologies including:

- * Fan-Out**
- * Fan-In**
- * Scatter-Gather**
- * Gather-Scatter**
- * Stack**
- * Cycle**
- * Bipartite**
- * Random**

Selecting any pattern dynamically filters the remaining visuals, enabling focused investigation of specific transaction behaviours.

A detailed investigation log presents each detected laundering pattern together with its transaction count, duration and start/end dates. This provides analysts with operational visibility into how suspicious transaction sequences evolve over time.

Complementing the investigation log are two comparative visualisations showing:

- * Transaction Volume by Laundering Pattern**
- * Pattern Duration by Laundering Type**

These visuals enable investigators to identify which laundering structures process the largest number of transactions and which remain active for the longest periods. Such information can support prioritisation strategies, investigative planning and resource allocation.

Unlike traditional AML dashboards that focus solely on account-level risk, this dashboard introduces a network-oriented perspective by highlighting the behavioural structures through which illicit funds may be dispersed across multiple accounts.

Together with the Executive Overview dashboard, it provides a comprehensive view of both who is considered high risk and how suspicious financial activity is conducted within the transaction network.

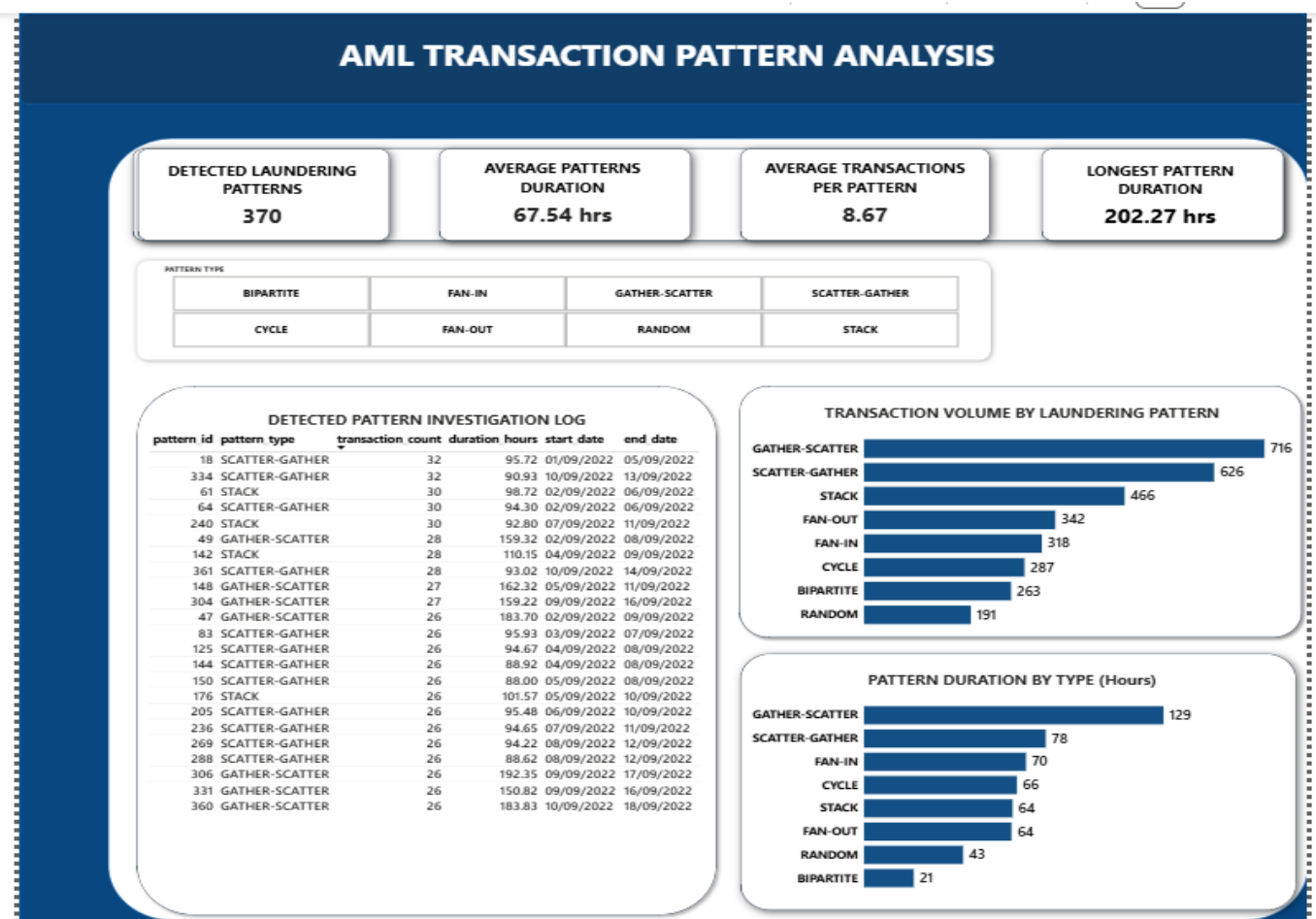


Fig.4: AML Transaction Pattern Analysis Dashboard

Dashboard Summary:

Together, the two Power BI dashboards translate complex analytical outputs into an accessible decision-support system for AML investigations. The Executive Overview dashboard provides strategic visibility into account-level risk, while the AML Transaction Pattern Analysis dashboard offers deeper insight into suspicious transaction structures and laundering typologies.

By combining engineered behavioural features, network analytics, risk scoring, machine learning outputs and interactive business intelligence, the final solution enables investigators to identify high-risk accounts, understand the behaviours driving those risks and prioritise investigations more effectively. This integration demonstrates how data analytics and machine learning can be transformed into practical tools that support real-world anti-money laundering operations.

The analytical pipeline developed in this project revealed that suspicious financial activity is concentrated within a relatively small proportion of monitored accounts. Although the majority of accounts exhibited normal transactional behaviour, a limited number demonstrated behavioural characteristics that consistently aligned with known money laundering typologies.

The risk scoring framework showed that transaction volume alone is not sufficient to identify suspicious behaviour. Accounts classified as High and Critical Risk typically displayed combinations of elevated transaction frequency, extensive fund dispersion through fan-out relationships, concentrated inbound transactions, large cumulative transaction values and confirmed laundering activity. These behavioural indicators proved to be more informative when evaluated collectively than when considered independently.

The network analysis further demonstrated that suspicious accounts rarely operate in isolation. High-risk accounts frequently formed interconnected transaction structures characterised by fan-in, fan-out, scatter-gather and stack patterns, suggesting coordinated movement of funds rather than isolated transactions. This reinforces the importance of analysing relationships between accounts instead of relying solely on transaction-level rules.

Model evaluation also highlighted the challenges associated with highly imbalanced AML datasets. Traditional machine learning models achieved high accuracy by predicting the majority class but performed poorly at identifying laundering accounts. After applying SMOTE and comparing multiple algorithms, the XGBoost model provided the best balance between precision, recall and F1-score, making it the most suitable model for supporting AML investigations within this project.

Finally, the integration of Python and Power BI demonstrated how advanced analytical outputs can be translated into interactive dashboards that support operational decision-making. Rather than presenting isolated metrics, the dashboards provide investigators with a prioritised view of high-risk accounts, behavioural drivers and transaction patterns, enabling faster and more informed investigative decisions.

Based on the findings of this project, several recommendations can enhance the effectiveness of anti-money laundering monitoring systems.

Financial institutions should adopt behavioural risk scoring models that evaluate multiple indicators simultaneously rather than relying solely on fixed transaction thresholds. Combining transaction activity, network behaviour and laundering history provides a more comprehensive assessment of customer risk.

Network analytics should become an integral component of AML monitoring programmes. Detecting patterns such as fan-in, fan-out, scatter-gather and cyclic fund movements can reveal organised laundering activities that may remain undetected using conventional rule-based systems.

Given the severe class imbalance commonly observed in AML datasets, machine learning models should be trained using appropriate resampling techniques such as SMOTE or alternative imbalance-handling methods. This improves the model's ability to identify suspicious accounts while reducing the bias toward legitimate transactions.

The Power BI dashboards developed in this project can also be integrated into compliance workflows to support continuous monitoring. Interactive visualisations allow investigators to prioritise alerts, understand the behavioural drivers behind elevated risk scores and focus investigative resources on the highest-risk accounts.

Finally, risk scoring frameworks should be reviewed and recalibrated periodically to reflect evolving laundering techniques, regulatory expectations and emerging financial crime trends.

Although the proposed framework produced meaningful analytical results, several limitations should be acknowledged.

The analysis was performed using a synthetic dataset developed for research purposes. While the dataset realistically simulates financial transactions and laundering behaviour, it cannot fully capture the complexity, diversity and unpredictability of real-world banking environments.

The engineered dataset represents account-level behavioural summaries over a fixed observation period. As a result, the analysis does not consider long-term behavioural changes or seasonal transaction patterns that may influence customer risk.

The risk scoring framework is based on predefined business rules and weighted behavioural indicators. Although these weights were designed to reflect realistic AML scenarios, they were manually defined and may differ from the scoring methodologies used by individual financial institutions.

Similarly, the machine learning models were evaluated using labelled synthetic data. Their performance may vary when applied to production banking data containing different customer behaviours, transaction volumes and regulatory requirements.

Finally, while the dashboards provide valuable decision-support capabilities, they are intended to assist investigators rather than replace expert judgement. Final AML decisions should always incorporate regulatory guidelines, investigative procedures and additional contextual information.

Several opportunities exist to extend this project beyond its current scope.

Future work could incorporate graph databases such as Neo4j to perform more advanced network analysis and identify complex relationships between customers, accounts and intermediary entities.

Real-time transaction streaming technologies could also be integrated to enable continuous monitoring and immediate detection of suspicious activities as transactions occur.

The machine learning framework could be expanded by evaluating additional ensemble and deep learning models, as well as anomaly detection techniques capable of identifying previously unseen laundering behaviours without relying solely on labelled data.

Further improvements may include incorporating customer demographic information, account profiles, geographic data and historical behavioural trends to develop more comprehensive risk assessment models.

The Power BI solution could also be enhanced through automated data refresh, drill-through investigation pages, role-based security and integration with enterprise AML platforms, allowing investigators to move seamlessly from executive monitoring to detailed case analysis.

This project demonstrates how data analytics, network analysis, machine learning and business intelligence can be integrated into a unified framework for anti-money laundering risk monitoring.

Starting with over five million transaction records, the project transformed raw financial data into an engineered account-level dataset that captures behavioural characteristics associated with suspicious financial activity. Through feature engineering, network analytics and a custom risk scoring framework, high-risk accounts were systematically identified based on their transactional behaviour rather than relying solely on traditional threshold-based rules.

Multiple machine learning models were developed and evaluated to address the challenges of detecting laundering activity within a highly imbalanced dataset. Comparative evaluation showed that the SMOTE-enhanced XGBoost model provided the strongest overall performance, demonstrating the importance of combining resampling techniques with robust classification algorithms in AML applications.

The analytical outputs were ultimately translated into two interactive Power BI dashboards: the Executive Overview of AML Risk Monitoring, which provides a strategic view of institutional risk exposure, and the AML Transaction Pattern Analysis dashboard, which enables investigators to explore suspicious transaction structures and laundering typologies in greater detail. Together, these dashboards transform complex analytical findings into actionable intelligence that supports risk-based decision-making.

Overall, this project illustrates the value of combining Python, machine learning, network analytics and Power BI to build an end-to-end AML monitoring solution. Beyond demonstrating technical proficiency, it reflects a practical approach to solving real-world financial crime challenges and showcases the application of modern data analytics techniques in supporting more effective anti-money laundering investigations.